Reverse dosimetry report

First morning urine (FMU) samples were collected from 83 individuals selected from the INSEF-ExpoQuim survey (Gilles et al., 2022; Govarts et al., 2023) between May 2019-2020. Sample were analysed for the quantification of multiple mycotoxins, including tenuazonic acid (TA) and deoxynivalenol, that showed to be the most occurent mycotoxins in the cohort. The current script aims to analyse the risk connected with DON and TA exposure in terms of HQ based on the defined tollerable daily intake (TDI) and threshold of toxicological concert (TTC), respectively. The script uses urinary concentrations expressed both as ng/mL and ng/mg creatinine (crea). The risk is calculate considering the FMU both as representative for the daily excretion (FMU approach) or as fraction of the daily excretion (24h approach).

Data

This chunk loads the dataset and applies the body-weight imputation for records with missing weight values. Modify "name.csv".

```
data <- read_csv("name.csv") df <- as.data.frame(data)
missing_bw <- is.na(df$weight)

df$weight[missing_bw] <- sapply(which(missing_bw), function(i) {

  sex_i <- df$sex[i]
  pool <- df$weight[df$sex == sex_i & !is.na(df$weight)]
  sample(pool, 1, replace = TRUE)
})

head(data)
```

```
## # A tibble: 6 x 74
##   ID     samplingseason   sex country nuts1 nuts2 degurba ageyears isced occ_s
##   <chr>           <dbl> <dbl> <chr>   <chr> <dbl>   <dbl>    <dbl> <dbl> <dbl>
## 1 EQ1003              3     0 PT      PT1       1       2       34     1     1
## 2 EQ1004              3     1 PT      PT1       1       2       38     2     1
## 3 EQ1005              3     0 PT      PT1       1       2       30     3    NA
## 4 EQ1007              3     1 PT      PT1       1       2       34     2     1
## 5 EQ1008              3     0 PT      PT1       1       1       39     3     1
## 6 EQ1009              3     1 PT      PT1       1       3       38     2     1
## # i 64 more variables: creatinine <dbl>, DON <dbl>, LOD <dbl>, LOQ <dbl>,
## #   don_imputed_results <dbl>, don_imputed_log <dbl>,
## #   don_imputed_creatinine <dbl>, don_imputed_creatinine_log <dbl>, TA <dbl>,
## #   TA_LOD <dbl>, TA_LOQ <dbl>, ta_imputed_results <dbl>, ta_imputed_log <dbl>,
## #   ta_imputed_creatinine <dbl>, ta_imputed_creatinine_log <dbl>,
## #   isco_2_digits <dbl>, isco_3_digits <dbl>, isco_4_digits <dbl>,
## #   income_hh <dbl>, vegetarian_fd <dbl>, dietgluten <dbl>, meat_fd <dbl>, ...
```

```r
colnames(data)
```

```
##  [1] "ID"                          "samplingseason"
##  [3] "sex"                         "country"
##  [5] "nuts1"                       "nuts2"
##  [7] "degurba"                     "ageyears"
##  [9] "isced"                       "occ_s"
## [11] "creatinine"                  "DON"
## [13] "LOD"                         "LOQ"
## [15] "don_imputed_results"         "don_imputed_log"
## [17] "don_imputed_creatinine"      "don_imputed_creatinine_log"
## [19] "TA"                          "TA_LOD"
## [21] "TA_LOQ"                      "ta_imputed_results"
## [23] "ta_imputed_log"              "ta_imputed_creatinine"
## [25] "ta_imputed_creatinine_log"   "isco_2_digits"
## [27] "isco_3_digits"               "isco_4_digits"
## [29] "income_hh"                   "vegetarian_fd"
## [31] "dietgluten"                  "meat_fd"
## [33] "poultry_fd"                  "fruveg_fd"
## [35] "organ_meat_fd"               "rice_fd"
## [37] "hazel_fd"                    "butter_fd"
## [39] "bread_fd"                    "cookie_fd"
## [41] "cheese_fd"                   "egg_fd"
## [43] "fat_fd"                      "popcorn_fd"
## [45] "smoke_fd"                    "flame_fd"
## [47] "potato_fried_fd"             "milk_drk"
## [49] "drink_wtr"                   "can_drk"
## [51] "bottle_drk"                  "beer"
## [53] "alcohol"                     "smoking"
## [55] "smoking_cigday"              "smoking_passive"
## [57] "meat_fd_2"                   "poultry_fd_2"
## [59] "fruveg_fd_2"                 "organ_meat_fd_2"
## [61] "rice_fd_2"                   "hazel_fd_2"
## [63] "butter_fd_2"                 "bread_fd_2"
## [65] "cookie_fd_2"                 "cheese_fd_2"
## [67] "egg_fd_2"                    "fat_fd_2"
## [69] "popcorn_fd_2"                "potato_fried_fd_2"
## [71] "milk_drk_2"                  "beer_2"
## [73] "height"                      "weight"
```

Parameters

This chunk defines the simulation settings, risk thresholds, plotting colours, and the food-frequency variables used later in the correlation analysis.

```
## [1] 1500
```

```
## [1] 1000
```

```
##  [1] "vegetarian_fd"   "meat_fd"         "poultry_fd"      "fruveg_fd"
##  [5] "organ_meat_fd"   "rice_fd"         "hazel_fd"        "butter_fd"
##  [9] "bread_fd"        "cookie_fd"       "cheese_fd"       "egg_fd"
```

```
## [13] "fat_fd"          "popcorn_fd"      "potato_fried_fd" "milk_drk"
## [17] "beer"            "alcohol"
```

```
## $`Meat and animal foods`
## [1] "meat_fd"       "poultry_fd"    "organ_meat_fd" "egg_fd"
##
## $`Cereal-based foods`
## [1] "rice_fd"    "bread_fd"   "cookie_fd"  "popcorn_fd"
##
## $Dairy
## [1] "milk_drk"  "cheese_fd" "butter_fd" "fat_fd"
##
## $`Plant-based foods`
## [1] "fruveg_fd"     "hazel_fd"      "vegetarian_fd"
##
## $`Fried and smoked foods`
## [1] "potato_fried_fd" "smoke_fd"        "flame_fd"
##
## $`Alcoholic drinks`
## [1] "beer"    "alcohol"
```

Helper functions

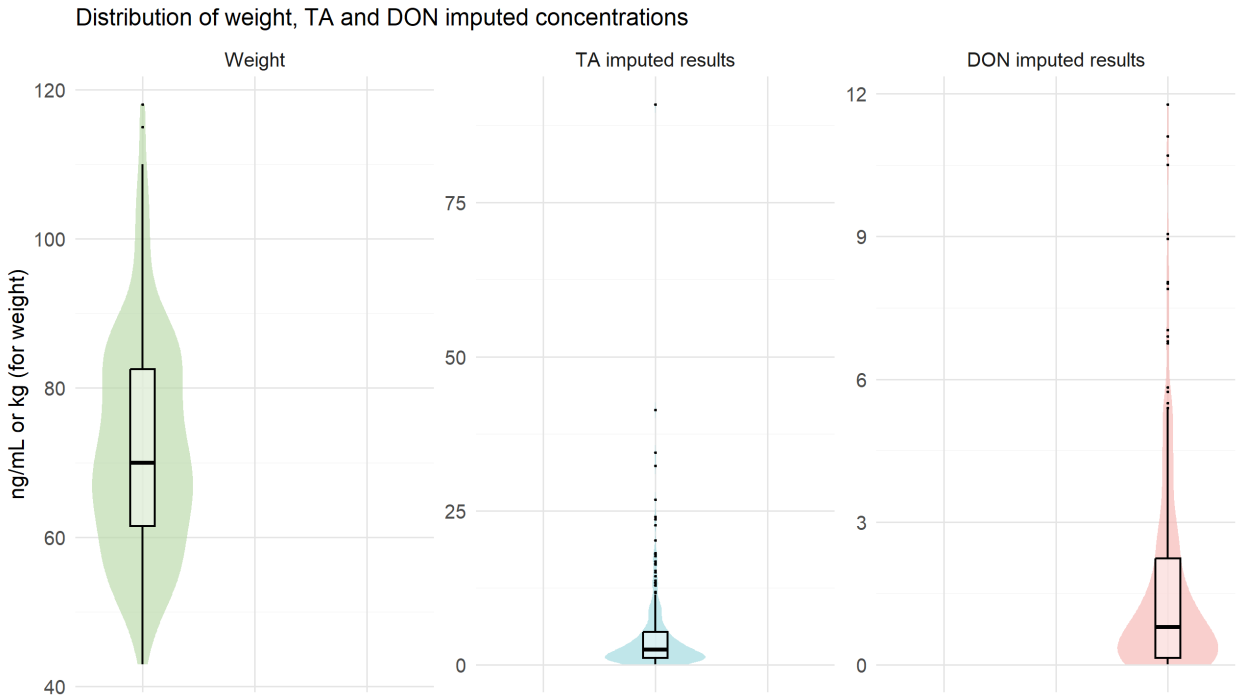
This chunk defines the plotting helper and the simulation functions used for the FMU, 24h, and creatinine-adjusted approaches.

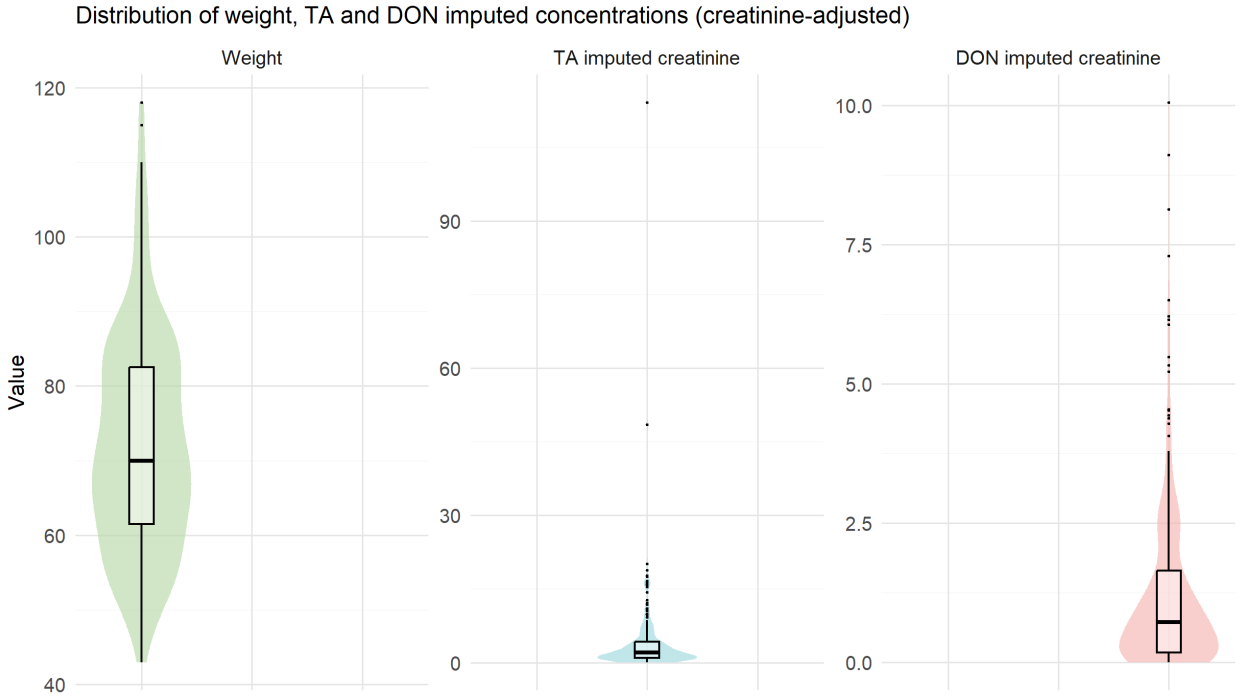
Simulation functions for input distributions and calculation of EDI

This chunk defines the simulation functions used for the fraction excreted in urine (FUE), first morning urine volume (FMU), 24h urinary volume (24h), the estimated daily intake (edi) and their creatinine-adjusted equivalents (_crea).

Input distributions

These chunks plot the input distributions of the two mycotoxins and the weight (after imputation of the NA), separately for the concentration expressed as ng/mL and ng/mg creatinine.





Monte Carlo simulation

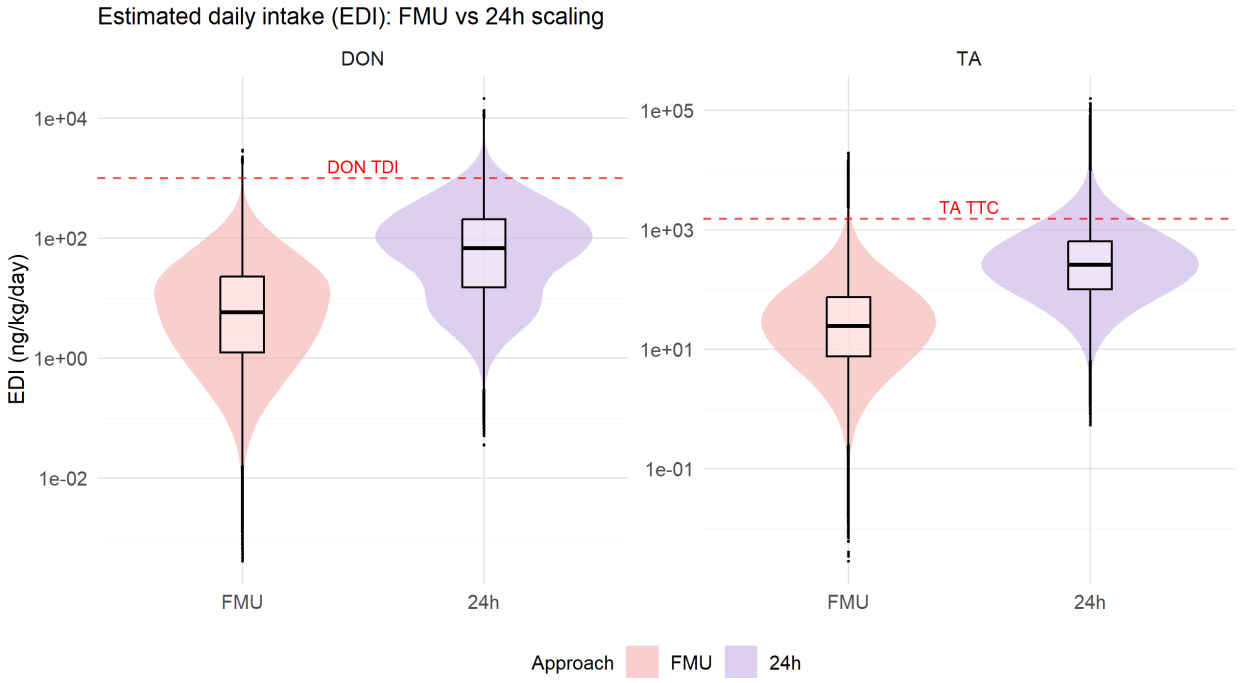
This chunck simulates the EDI and HQ using 4 different approachs for the two mycotoxins. The approaches differ for the urinary volume and the concentration used. The FMU simulated is used as representative for the whole day or as fraction of the 24h urinary excretion. The concentration is expressed as ng/mL (volume) or as ng/mg creatinine (crea).

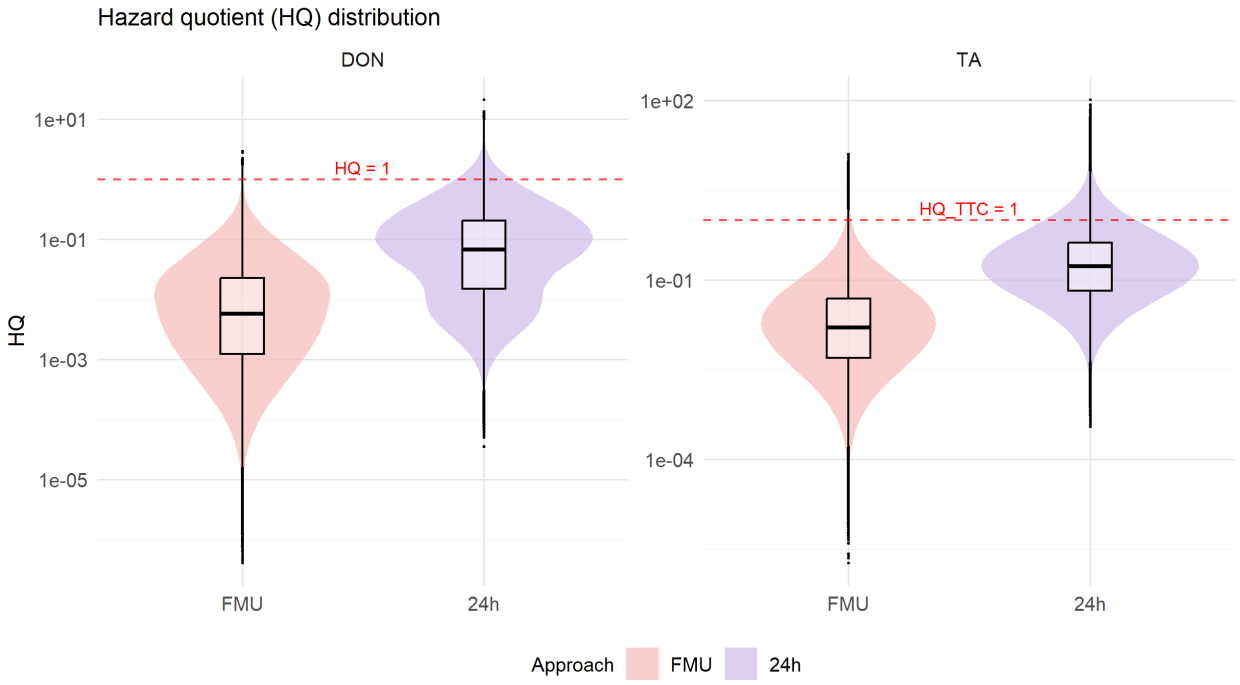
```
## # A tibble: 8 x 10
##   Compound Approach Median_EDI Mean_EDI P95_EDI Median_HQ Mean_HQ P95_HQ
##   <chr>    <fct>         <dbl>    <dbl>   <dbl>     <dbl>   <dbl>  <dbl>
```

```
## 1 DON      Crea 24h      28.4      56.2   206.    0.0284   0.0562 0.206
## 2 DON      Crea FMU       2.28     10.4    45.6   0.00228  0.0104 0.0456
## 3 DON      FMU            5.83     24.7   110.    0.00583  0.0247 0.110
## 4 DON      24h           67.4     189.    780.    0.0674   0.189  0.780
## 5 TA       Crea 24h      98.4     216.    733.    0.0656   0.144  0.489
## 6 TA       Crea FMU       9.55     40.1   157.    0.00636  0.0268 0.105
## 7 TA       FMU           24.4      82.5   329.    0.0163   0.0550 0.219
## 8 TA       24h          253.      632.   2337.    0.169    0.421  1.56
## # i 2 more variables: Prob_HQ_above_1 <dbl>, Prob_HQ_above_1_percent <dbl>
```

Volume-based plots

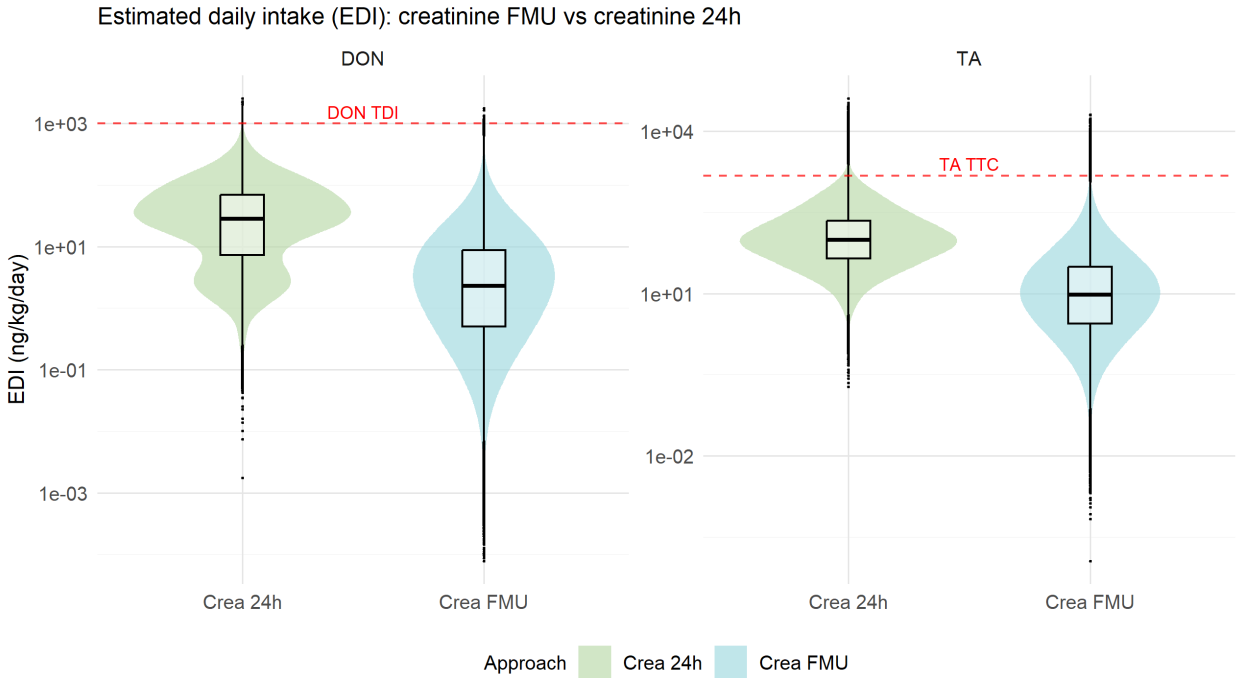
These chuncks plot the EDI and HQ obtained for the FMU and 24h approaches using the concentration expressed in ng/mL.

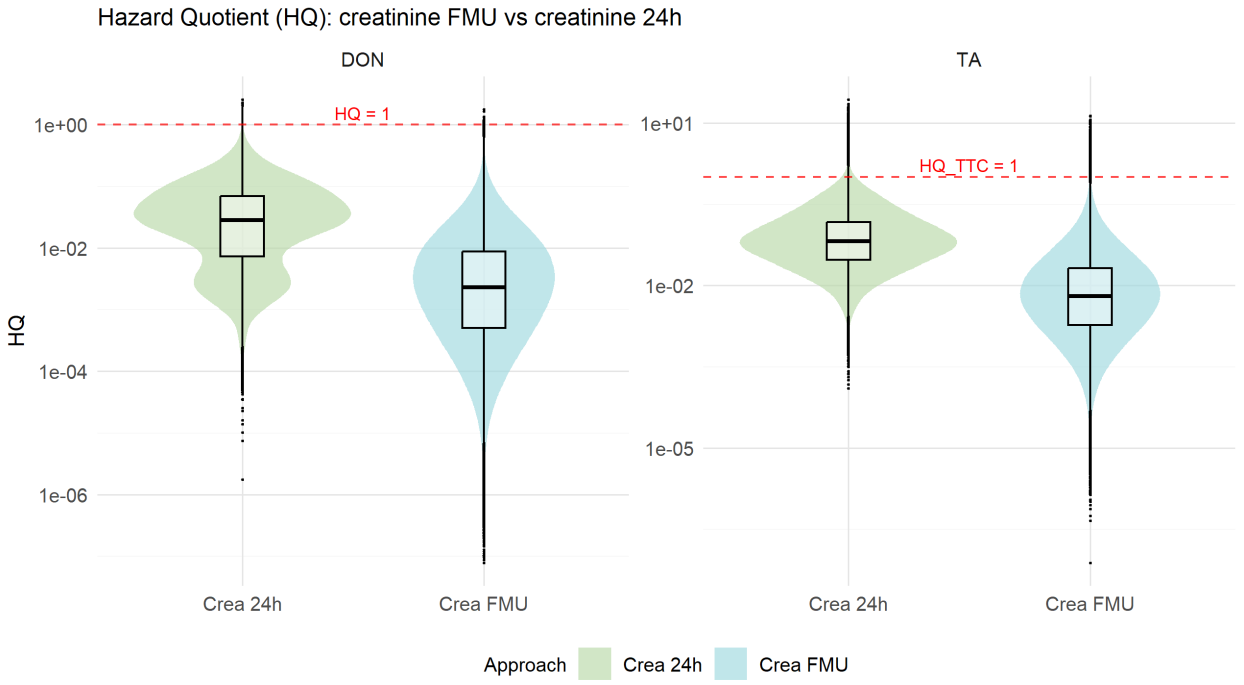




Creatinine-based plots

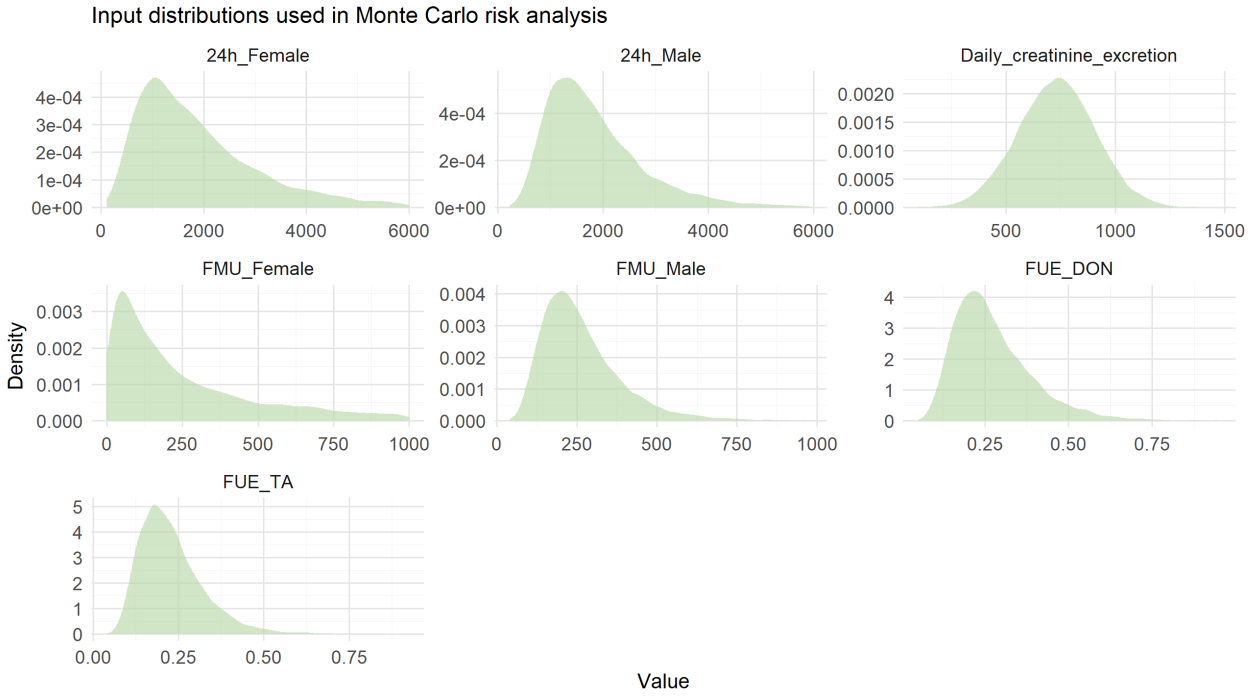
These chuncks plot the EDI and HQ obtained for the FMU and 24h approaches using the concentration expressed in ng/mg creatinine.





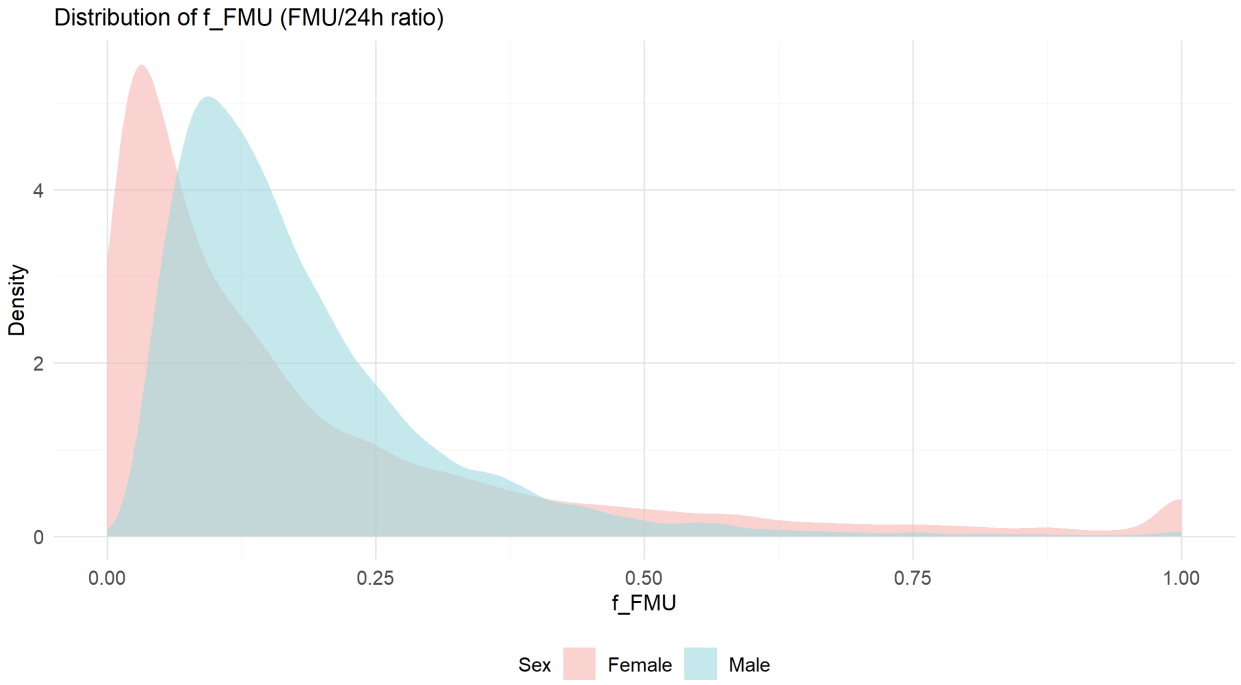
Simulated input distributions used in Monte Carlo

This chunck plots the previously simulated input distributions used in the Monte Carlo risk analysis.



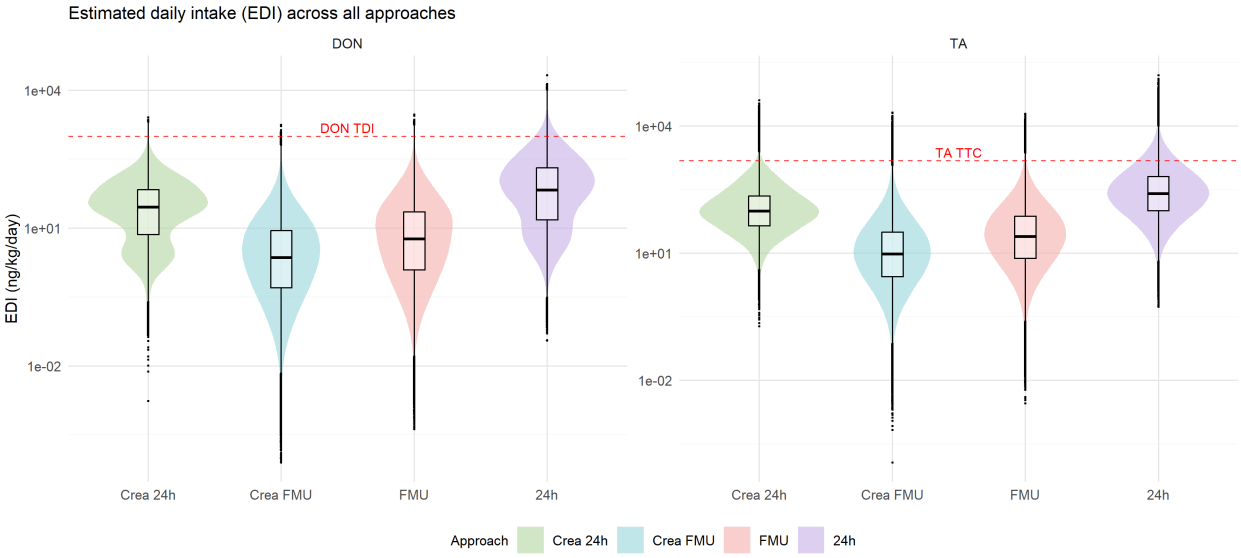
Distribution of f_FMU (FMU/24h ratio)

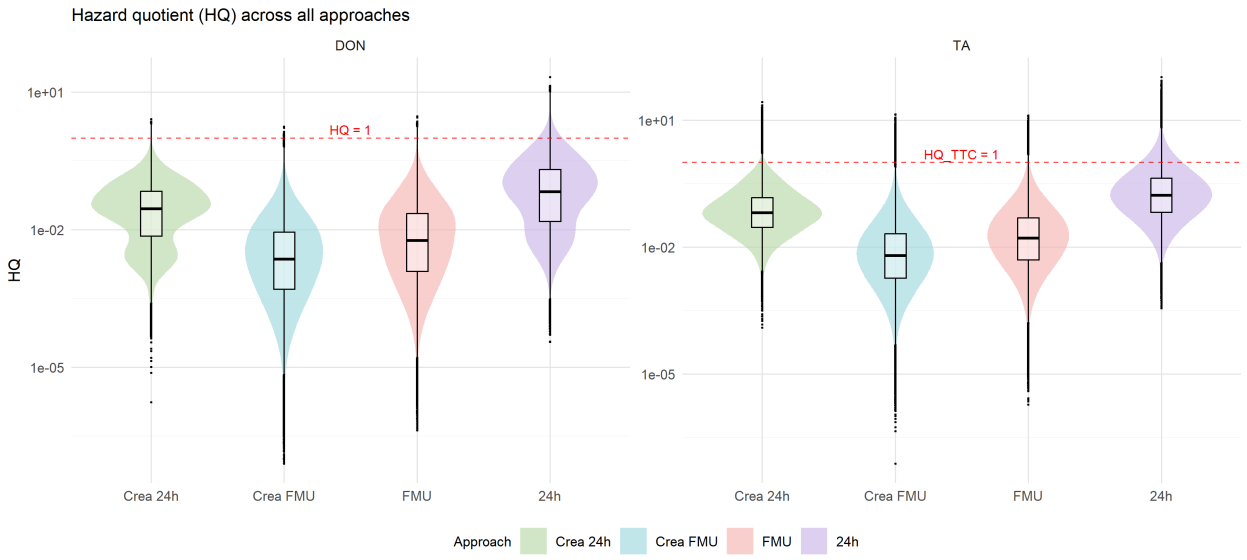
This chunck plots the f_FMU (FMU/24h ratio) for males and females.



Comparison across all four approaches

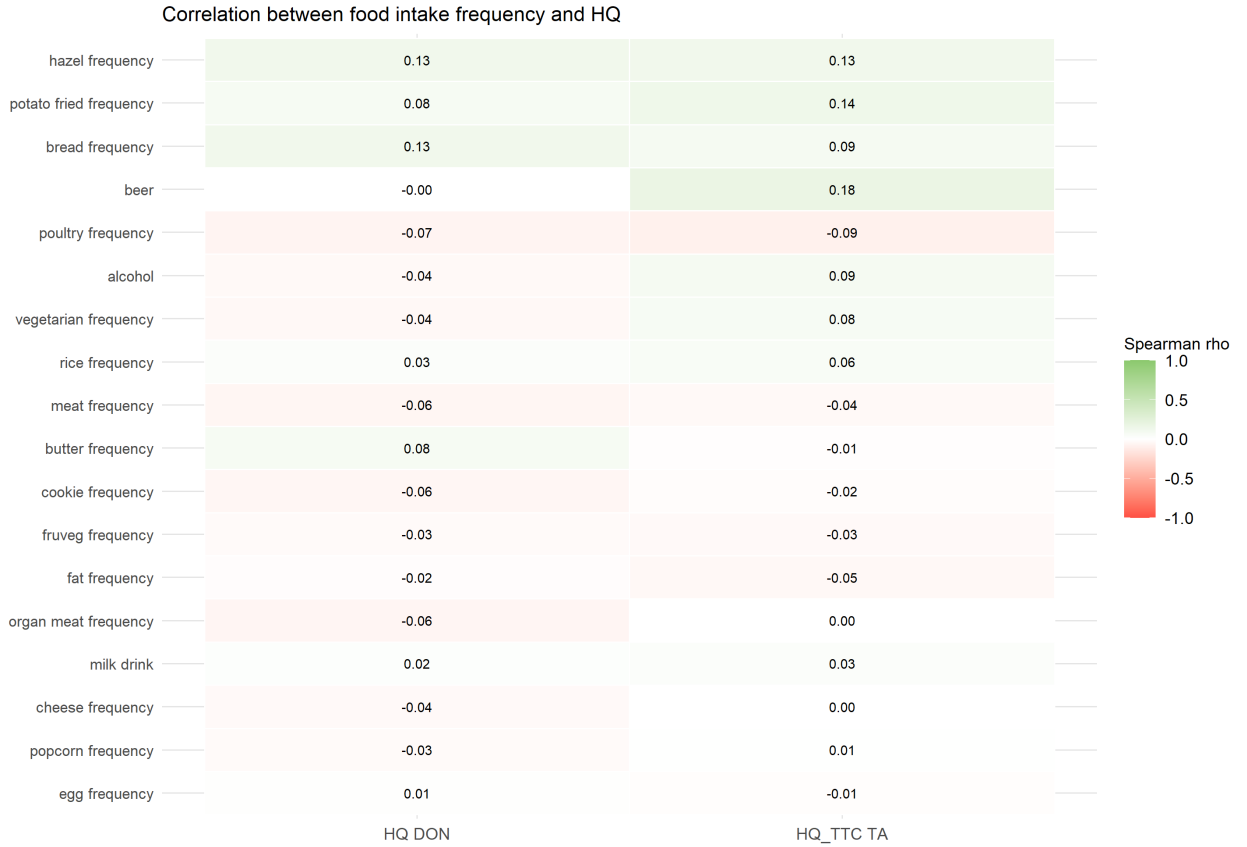
These chucks create a plot for the comparison of the EDI and HQ obtained with the 4 different approaches.





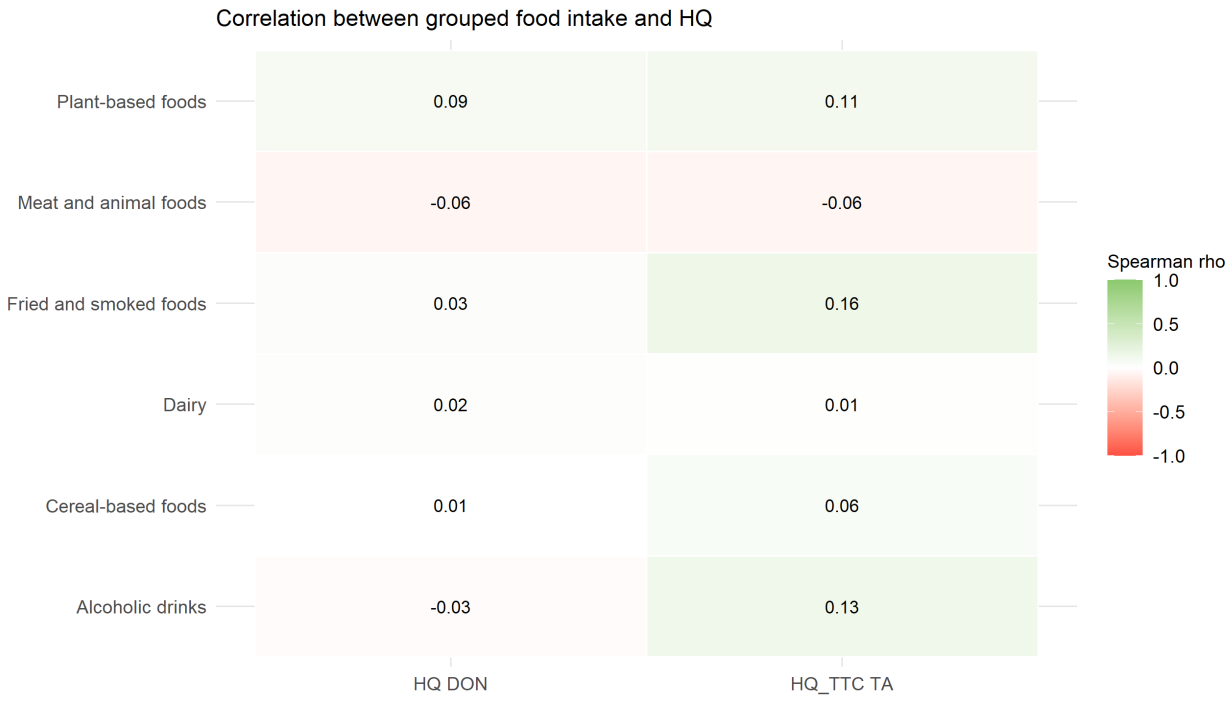
Correlation between food intake and HQ

This chunk summarises each participant's median HQ for the volume-based 24h approach for TA and DON, and then calculates the correlations with the main food-frequency variables and adjusted p-values. Positive correlations indicate that higher intake frequency tends to be associated with higher HQ. Since food intake variables are ordinal or semi-quantitative frequency variables, not continuous measurements, Spearman correlation was used.



Correlation between grouped food intake and HQ

This chunk combines related foods into broader food groups, then calculates Spearman correlations and adjusted p-values for the grouped scores against HQ for TA and DON.



At-risk subgroups

This chunk recodes the main demographic and anthropometric variables and identifies potentially at-risk subgroups based on the median HQ from the volume-based 24h approach. For categorical variables, distribution differences are tested between groups with Wilcoxon or Kruskal-Wallis tests. For age, Spearman correlation is used.



Adjusted regression models for HQ

This chunk fits adjusted regression models for each mycotoxin using the same subgroup variables considered above. Linear regression is applied to `log(HQ)`, while logistic regression is applied to the binary outcome `HQ > 1`.

Interpretation

The linear regression model estimates how each variable is associated with an increased of decresed HQ. For example, an estimate = 0.69 ($\exp(0.69) = 2.0$) means about 2 times higher HQ, while an estimate = -0.69 means about 2 times lower HQ. The logistic regression model estimates how each variable is associated with the odds of exceeding the risk threshold. For example, an estimate = 1.10 for age, means that each one-unit increase in age increases the odds by 10%. Coefficients equal to zero or extremely high should be interpreted as unstable and unreliable, due to a low sample size for the specific subgroup.

The p-values are the significance test for the coefficients. The null hypothesis is represented by a coefficient = 0, meaning no association with log(HQ) or with odds of HQ >1.

```
## [1] "Results linear regression"
```

```
##                          term     estimate  std_error   statistic
## 1                 (Intercept) -0.124482562 0.96688412 -0.12874610
## 2                    ageyears -0.019769190 0.02181534 -0.90620600
## 3                sex_groupMale -0.284321744 0.15425092 -1.84324184
```

```
## 4               isced_groupMedium -0.401957052 0.19418202 -2.07000140
## 5                 isced_groupHigh -0.554509744 0.19523393 -2.84023243
## 6        degurba_groupRural areas -0.094285958 0.21016924 -0.44861920
## 7  degurba_groupTowns and suburbs -0.124264377 0.20820361 -0.59684063
## 8              nuts2_groupNUTS2 2  0.119301253 0.28820246  0.41394946
## 9              nuts2_groupNUTS2 3  0.317840881 0.25586567  1.24221778
## 10             nuts2_groupNUTS2 4 -0.083900213 0.32066514 -0.26164432
## 11             nuts2_groupNUTS2 5  0.419059034 0.26602498  1.57526197
## 12             nuts2_groupNUTS2 6  0.218562686 0.23944210  0.91279975
## 13             nuts2_groupNUTS2 7  0.014498724 0.27216363  0.05327208
## 14             occ_groupocc_s = 1 -0.295562456 0.27330339 -1.08144453
## 15         bmi_groupNormal weight -0.112167755 0.53974673 -0.20781553
## 16            bmi_groupOverweight -0.337882628 0.54733384 -0.61732457
## 17               bmi_groupObesity -0.310964641 0.55973852 -0.55555340
## 18                    (Intercept) -1.230392336 1.34223720 -0.91667281
## 19                       ageyears -0.020199499 0.03028425 -0.66699691
## 20                  sex_groupMale -0.332174571 0.21413250 -1.55125711
## 21              isced_groupMedium  0.198008988 0.26956522  0.73454947
## 22                isced_groupHigh -0.351921812 0.27102549 -1.29848232
## 23       degurba_groupRural areas -0.165844456 0.29175882 -0.56842996
## 24 degurba_groupTowns and suburbs -0.234581620 0.28903012 -0.81161651
## 25             nuts2_groupNUTS2 2  0.603482685 0.40008524  1.50838528
## 26             nuts2_groupNUTS2 3 -0.470544047 0.35519501 -1.32474846
## 27             nuts2_groupNUTS2 4  0.001781242 0.44515022  0.00400144
## 28             nuts2_groupNUTS2 5  0.109859620 0.36929826  0.29748209
## 29             nuts2_groupNUTS2 6  0.633986020 0.33239566  1.90732338
## 30             nuts2_groupNUTS2 7  0.382794657 0.37781999  1.01316678
## 31             occ_groupocc_s = 1 -0.417234439 0.37940222 -1.09971535
## 32         bmi_groupNormal weight -0.208920751 0.74928125 -0.27882821
## 33            bmi_groupOverweight -0.374465717 0.75981374 -0.49283883
## 34               bmi_groupObesity -0.416745856 0.77703403 -0.53632896
##        p_value exp_estimate percent_change signif_label   HQ_type
## 1  0.897658087    0.8829537    -11.7046342              HQ_TTC TA
## 2  0.365665863    0.9804249     -1.9575061              HQ_TTC TA
## 3  0.066431841    0.7525245    -24.7475514              HQ_TTC TA
## 4  0.039439715    0.6690095    -33.0990522            * HQ_TTC TA
## 5  0.004864649    0.5743538    -42.5646228           ** HQ_TTC TA
## 6  0.654080032    0.9100225     -8.9977503              HQ_TTC TA
## 7  0.551133134    0.8831463    -11.6853674              HQ_TTC TA
## 8  0.679252387    1.1267093     12.6709291              HQ_TTC TA
## 9  0.215275504    1.3741576     37.4157590              HQ_TTC TA
## 10 0.793802820    0.9195230     -8.0476992              HQ_TTC TA
## 11 0.116411963    1.5205301     52.0530115              HQ_TTC TA
## 12 0.362193485    1.2442870     24.4287014              HQ_TTC TA
## 13 0.957556030    1.0146043      1.4604340              HQ_TTC TA
## 14 0.280501450    0.7441129    -25.5887061              HQ_TTC TA
## 15 0.835535690    0.8938943    -10.6105711              HQ_TTC TA
## 16 0.537560916    0.7132790    -28.6720998              HQ_TTC TA
## 17 0.578993898    0.7327398    -26.7260215              HQ_TTC TA
## 18 0.360163591    0.2921779    -70.7822077                 HQ DON
## 19 0.505365604    0.9800031     -1.9996856                 HQ DON
## 20 0.122056435    0.7173621    -28.2637919                 HQ DON
## 21 0.463275883    1.2189733     21.8973349                 HQ DON
## 22 0.195272171    0.7033351    -29.6664888                 HQ DON
```

```
## 23 0.570233667    0.8471780    -15.2822003                HQ DON
## 24 0.417754110    0.7909017    -20.9098322                HQ DON
## 25 0.132670107    1.8284757     82.8475729                HQ DON
## 26 0.186417749    0.6246623    -37.5337670                HQ DON
## 27 0.996810390    1.0017828      0.1782829                HQ DON
## 28 0.766336026    1.1161214     11.6121378                HQ DON
## 29 0.057580633    1.8851097     88.5109708                HQ DON
## 30 0.311922421    1.4663769     46.6376889                HQ DON
## 31 0.272473359    0.6588664    -34.1133562                HQ DON
## 32 0.780598307    0.8114595    -18.8540459                HQ DON
## 33 0.622542421    0.6876566    -31.2343416                HQ DON
## 34 0.592189787    0.6591884    -34.0811573                HQ DON
##                           model
## 1  Linear regression on log(HQ)
## 2  Linear regression on log(HQ)
## 3  Linear regression on log(HQ)
## 4  Linear regression on log(HQ)
## 5  Linear regression on log(HQ)
## 6  Linear regression on log(HQ)
## 7  Linear regression on log(HQ)
## 8  Linear regression on log(HQ)
## 9  Linear regression on log(HQ)
## 10 Linear regression on log(HQ)
## 11 Linear regression on log(HQ)
## 12 Linear regression on log(HQ)
## 13 Linear regression on log(HQ)
## 14 Linear regression on log(HQ)
## 15 Linear regression on log(HQ)
## 16 Linear regression on log(HQ)
## 17 Linear regression on log(HQ)
## 18 Linear regression on log(HQ)
## 19 Linear regression on log(HQ)
## 20 Linear regression on log(HQ)
## 21 Linear regression on log(HQ)
## 22 Linear regression on log(HQ)
## 23 Linear regression on log(HQ)
## 24 Linear regression on log(HQ)
## 25 Linear regression on log(HQ)
## 26 Linear regression on log(HQ)
## 27 Linear regression on log(HQ)
## 28 Linear regression on log(HQ)
## 29 Linear regression on log(HQ)
## 30 Linear regression on log(HQ)
## 31 Linear regression on log(HQ)
## 32 Linear regression on log(HQ)
## 33 Linear regression on log(HQ)
## 34 Linear regression on log(HQ)

## [1] "Results logistic regression"

##                            term     estimate    std_error     statistic
## 1                   (Intercept) 3.789773e-02 3.484661e+00 -9.392202e-01
## 2                      ageyears 1.084460e+00 8.911613e-02  9.098513e-01
## 3                 sex_groupMale 9.809644e-01 5.829899e-01 -3.296639e-02
```

```
## 4               isced_groupMedium 7.396400e-01 6.164666e-01 -4.892263e-01
## 5                 isced_groupHigh 2.009810e-01 8.073905e-01 -1.987322e+00
## 6        degurba_groupRural areas 1.930416e+00 7.800969e-01  8.431457e-01
## 7  degurba_groupTowns and suburbs 8.012254e-01 8.439601e-01 -2.625871e-01
## 8              nuts2_groupNUTS2 2 4.777611e-08 1.114338e+03 -1.512713e-02
## 9              nuts2_groupNUTS2 3 1.147011e-01 1.185424e+00 -1.826710e+00
## 10             nuts2_groupNUTS2 4 4.179812e-01 1.296080e+00 -6.730440e-01
## 11             nuts2_groupNUTS2 5 5.557311e-01 8.632036e-01 -6.805703e-01
## 12             nuts2_groupNUTS2 6 5.620481e-01 7.522670e-01 -7.659087e-01
## 13             nuts2_groupNUTS2 7 3.762759e-01 9.314733e-01 -1.049341e+00
## 14             occ_groupocc_s = 1 2.026316e+00 1.333798e+00  5.294799e-01
## 15         bmi_groupNormal weight 2.164483e-01 1.517636e+00 -1.008412e+00
## 16            bmi_groupOverweight 1.245057e-01 1.555244e+00 -1.339599e+00
## 17               bmi_groupObesity 1.955499e-01 1.567354e+00 -1.041207e+00
## 18                    (Intercept) 2.352818e-21 1.195234e+05 -3.974005e-04
## 19                       ageyears 3.069377e-01 9.557785e-01 -1.235758e+00
## 20                  sex_groupMale 7.500145e+06 1.460775e+04  1.083701e-03
## 21              isced_groupMedium 1.234127e-15 2.708709e+04 -1.267335e-03
## 22                isced_groupHigh 7.288478e+05 2.040282e+04  6.616350e-04
## 23       degurba_groupRural areas 7.007151e-21 1.600534e+04 -2.899491e-03
## 24 degurba_groupTowns and suburbs 3.571726e-28 2.343477e+04 -2.696820e-03
## 25             nuts2_groupNUTS2 2 9.342668e+01 4.994855e+04  9.083702e-05
## 26             nuts2_groupNUTS2 3 4.883619e-01 4.464711e+04 -1.605252e-05
## 27             nuts2_groupNUTS2 4 3.378536e-01 5.938652e+04 -1.827254e-05
## 28             nuts2_groupNUTS2 5 3.384152e+30 3.826599e+04  1.837053e-03
## 29             nuts2_groupNUTS2 6 2.610447e+21 3.592713e+04  1.372607e-03
## 30             nuts2_groupNUTS2 7 1.060871e+00 4.705195e+04  1.255846e-06
## 31             occ_groupocc_s = 1 4.691417e-04 6.915085e+00 -1.108389e+00
## 32         bmi_groupNormal weight 1.316891e+16 1.152424e+05  3.220744e-04
## 33            bmi_groupOverweight 7.150482e+06 1.148825e+05  1.373812e-04
## 34               bmi_groupObesity 8.204193e-05 1.178744e+05 -7.981612e-05
##      p_value percent_change signif_label   HQ_type
## 1  0.3476177  -9.621023e+01              HQ_TTC TA
## 2  0.3629010   8.446028e+00              HQ_TTC TA
## 3  0.9737014  -1.903557e+00              HQ_TTC TA
## 4  0.6246815  -2.603600e+01              HQ_TTC TA
## 5  0.0468867  -7.990190e+01            * HQ_TTC TA
## 6  0.3991470   9.304157e+01              HQ_TTC TA
## 7  0.7928689  -1.987746e+01              HQ_TTC TA
## 8  0.9879308  -1.000000e+02              HQ_TTC TA
## 9  0.0677434  -8.852989e+01              HQ_TTC TA
## 10 0.5009193  -5.820188e+01              HQ_TTC TA
## 11 0.4961434  -4.442689e+01              HQ_TTC TA
## 12 0.4437306  -4.379519e+01              HQ_TTC TA
## 13 0.2940214  -6.237241e+01              HQ_TTC TA
## 14 0.5964726   1.026316e+02              HQ_TTC TA
## 15 0.3132565  -7.835517e+01              HQ_TTC TA
## 16 0.1803758  -8.754943e+01              HQ_TTC TA
## 17 0.2977796  -8.044501e+01              HQ_TTC TA
## 18 0.9996829  -1.000000e+02                 HQ DON
## 19 0.2165487  -6.930623e+01                 HQ DON
## 20 0.9991353   7.500144e+08                 HQ DON
## 21 0.9989888  -1.000000e+02                 HQ DON
## 22 0.9994721   7.288468e+07                 HQ DON
```

```
## 23 0.9976865  -1.000000e+02                HQ DON
## 24 0.9978483  -1.000000e+02                HQ DON
## 25 0.9999275   9.242668e+03                HQ DON
## 26 0.9999872  -5.116381e+01                HQ DON
## 27 0.9999854  -6.621464e+01                HQ DON
## 28 0.9985342   3.384152e+32                HQ DON
## 29 0.9989048   2.610447e+23                HQ DON
## 30 0.9999990   6.087071e+00                HQ DON
## 31 0.2676937  -9.995309e+01                HQ DON
## 32 0.9997430   1.316891e+18                HQ DON
## 33 0.9998904   7.150481e+08                HQ DON
## 34 0.9999363  -9.999180e+01                HQ DON
##                            model
## 1  Logistic regression on HQ > 1
## 2  Logistic regression on HQ > 1
## 3  Logistic regression on HQ > 1
## 4  Logistic regression on HQ > 1
## 5  Logistic regression on HQ > 1
## 6  Logistic regression on HQ > 1
## 7  Logistic regression on HQ > 1
## 8  Logistic regression on HQ > 1
## 9  Logistic regression on HQ > 1
## 10 Logistic regression on HQ > 1
## 11 Logistic regression on HQ > 1
## 12 Logistic regression on HQ > 1
## 13 Logistic regression on HQ > 1
## 14 Logistic regression on HQ > 1
## 15 Logistic regression on HQ > 1
## 16 Logistic regression on HQ > 1
## 17 Logistic regression on HQ > 1
## 18 Logistic regression on HQ > 1
## 19 Logistic regression on HQ > 1
## 20 Logistic regression on HQ > 1
## 21 Logistic regression on HQ > 1
## 22 Logistic regression on HQ > 1
## 23 Logistic regression on HQ > 1
## 24 Logistic regression on HQ > 1
## 25 Logistic regression on HQ > 1
## 26 Logistic regression on HQ > 1
## 27 Logistic regression on HQ > 1
## 28 Logistic regression on HQ > 1
## 29 Logistic regression on HQ > 1
## 30 Logistic regression on HQ > 1
## 31 Logistic regression on HQ > 1
## 32 Logistic regression on HQ > 1
## 33 Logistic regression on HQ > 1
## 34 Logistic regression on HQ > 1
```

Summary tables

```
## # A tibble: 4 x 10
##   Compound Approach Median_EDI Mean_EDI P95_EDI Median_HQ Mean_HQ P95_HQ
##   <chr>    <fct>         <dbl>    <dbl>   <dbl>     <dbl>   <dbl>  <dbl>
```

```
## 1 DON      FMU            5.83     24.7    110.   0.00583  0.0247  0.110
## 2 DON      24h           67.4     189.     780.   0.0674   0.189   0.780
## 3 TA       FMU           24.4      82.5    329.   0.0163   0.0550  0.219
## 4 TA       24h          253.      632.    2337.   0.169    0.421   1.56
## # i 2 more variables: Prob_HQ_above_1 <dbl>, Prob_HQ_above_1_percent <dbl>
```

```
## # A tibble: 4 x 10
##   Compound Approach Median_EDI Mean_EDI P95_EDI Median_HQ Mean_HQ P95_HQ
##   <chr>    <fct>         <dbl>    <dbl>   <dbl>     <dbl>   <dbl>  <dbl>
## 1 DON      Crea 24h      28.4      56.2   206.    0.0284   0.0562 0.206
## 2 DON      Crea FMU       2.28     10.4    45.6   0.00228  0.0104 0.0456
## 3 TA       Crea 24h      98.4     216.    733.    0.0656   0.144  0.489
## 4 TA       Crea FMU       9.55     40.1   157.    0.00636  0.0268 0.105
## # i 2 more variables: Prob_HQ_above_1 <dbl>, Prob_HQ_above_1_percent <dbl>
```

```
## # A tibble: 8 x 10
##   Compound Approach Median_EDI Mean_EDI P95_EDI Median_HQ Mean_HQ P95_HQ
##   <chr>    <fct>         <dbl>    <dbl>   <dbl>     <dbl>   <dbl>  <dbl>
## 1 DON      Crea 24h      28.4      56.2   206.    0.0284   0.0562 0.206
## 2 DON      Crea FMU       2.28     10.4    45.6   0.00228  0.0104 0.0456
## 3 DON      FMU            5.83     24.7   110.    0.00583  0.0247 0.110
## 4 DON      24h           67.4     189.    780.    0.0674   0.189  0.780
## 5 TA       Crea 24h      98.4     216.    733.    0.0656   0.144  0.489
## 6 TA       Crea FMU       9.55     40.1   157.    0.00636  0.0268 0.105
## 7 TA       FMU           24.4      82.5   329.    0.0163   0.0550 0.219
## 8 TA       24h          253.      632.   2337.    0.169    0.421  1.56
## # i 2 more variables: Prob_HQ_above_1 <dbl>, Prob_HQ_above_1_percent <dbl>
```

```
## # A tibble: 36 x 5
##    HQ_type food_label             spearman_rho p_value p_value_adj
##    <chr>   <fct>                         <dbl>   <dbl>       <dbl>
##  1 HQ DON  hazel frequency             0.129   0.0326       0.240
##  2 HQ DON  bread frequency             0.128   0.0332       0.240
##  3 HQ DON  butter frequency            0.0777  0.198        0.663
##  4 HQ DON  potato fried frequency      0.0784  0.193        0.663
##  5 HQ DON  meat frequency             -0.0579  0.337        0.791
##  6 HQ DON  poultry frequency          -0.0651  0.280        0.791
##  7 HQ DON  organ meat frequency       -0.0596  0.323        0.791
##  8 HQ DON  cookie frequency           -0.0563  0.352        0.791
##  9 HQ DON  vegetarian frequency       -0.0443  0.463        0.866
## 10 HQ DON  rice frequency              0.0337  0.578        0.866
## # i 26 more rows
```

```
## # A tibble: 12 x 5
##    HQ_type   food_group             spearman_rho p_value p_value_adj
##    <chr>     <chr>                         <dbl>   <dbl>       <dbl>
##  1 HQ DON    Plant-based foods           0.0853  0.157        0.471
##  2 HQ DON    Meat and animal foods      -0.0609  0.313        0.604
##  3 HQ DON    Fried and smoked foods      0.0294  0.626        0.896
##  4 HQ DON    Alcoholic drinks           -0.0256  0.672        0.896
##  5 HQ DON    Dairy                       0.0183  0.762        0.907
##  6 HQ DON    Cereal-based foods          0.00543 0.928        0.928
##  7 HQ_TTC TA Fried and smoked foods      0.155   0.00973      0.117
```

```
##  8 HQ_TTC TA Alcoholic drinks            0.131   0.0290        0.174
##  9 HQ_TTC TA Plant-based foods           0.108   0.0734        0.293
## 10 HQ_TTC TA Meat and animal foods      -0.0561  0.352         0.604
## 11 HQ_TTC TA Cereal-based foods          0.0589  0.328         0.604
## 12 HQ_TTC TA Dairy                       0.0129  0.831         0.907
```

```
## # A tibble: 14 x 7
##    subgroup_variable   HQ_type method statistic p_value p_value_adj signif_label
##    <chr>               <chr>   <chr>      <dbl>   <dbl>       <dbl> <chr>
##  1 Sex                 HQ_TTC~ Wilco~   1.18e+4 0.111        0.332  ""
##  2 Sex                 HQ DON  Wilco~   1.14e+4 0.283        0.679  ""
##  3 Education level     HQ_TTC~ Krusk~   1.35e+1 0.00118      0.0142 "*"
##  4 Education level     HQ DON  Krusk~   1.01e+1 0.00631      0.0330 "*"
##  5 Degree of urbaniza~ HQ_TTC~ Krusk~   1.09e+0 0.580        0.696  ""
##  6 Degree of urbaniza~ HQ DON  Krusk~   1.09e+0 0.579        0.696  ""
##  7 NUTS2 region        HQ_TTC~ Krusk~   6.75e+0 0.344        0.689  ""
##  8 NUTS2 region        HQ DON  Krusk~   1.73e+1 0.00825      0.0330 "*"
##  9 Occupation          HQ_TTC~ Wilco~   2.86e+3 0.405        0.694  ""
## 10 Occupation          HQ DON  Wilco~   2.79e+3 0.519        0.696  ""
## 11 BMI group           HQ_TTC~ Krusk~   6.09e-1 0.894        0.894  ""
## 12 BMI group           HQ DON  Krusk~   8.34e-1 0.841        0.894  ""
## 13 Age                 HQ_TTC~ Spear~  -8.76e-2 0.133        0.267  ""
## 14 Age                 HQ DON  Spear~  -6.14e-2 0.293        0.293  ""
```

```
## # A tibble: 42 x 7
##    subgroup_variable    subgroup_level HQ_type     n median_high_HQ mean_high_HQ
##    <chr>                <chr>          <chr>   <int>          <dbl>        <dbl>
##  1 NUTS2 region         NUTS2 2        HQ DON     30         0.154        0.236
##  2 NUTS2 region         NUTS2 6        HQ DON     62         0.119        0.205
##  3 BMI group            Underweight    HQ DON      5         0.101        0.179
##  4 Education level      Low            HQ DON     59         0.0999       0.186
##  5 Education level      Medium         HQ DON    107         0.0925       0.154
##  6 Occupation           occ_s = 0      HQ DON     20         0.0911       0.178
##  7 Degree of urbanizat~ Cities         HQ DON     80         0.0869       0.161
##  8 Sex                  Female         HQ DON    124         0.0856       0.154
##  9 NUTS2 region         NUTS2 4        HQ DON     25         0.0799       0.138
## 10 BMI group            Obesity        HQ DON     44         0.0754       0.109
## # i 32 more rows
## # i 1 more variable: prob_exceed_1 <dbl>
```

```
##      HQ_type                        model                          term
## 1  HQ_TTC TA Linear regression on log(HQ)                   (Intercept)
## 2  HQ_TTC TA Linear regression on log(HQ)                      ageyears
## 3  HQ_TTC TA Linear regression on log(HQ)                  sex_groupMale
## 4  HQ_TTC TA Linear regression on log(HQ)               isced_groupMedium
## 5  HQ_TTC TA Linear regression on log(HQ)                 isced_groupHigh
## 6  HQ_TTC TA Linear regression on log(HQ)       degurba_groupRural areas
## 7  HQ_TTC TA Linear regression on log(HQ) degurba_groupTowns and suburbs
## 8  HQ_TTC TA Linear regression on log(HQ)              nuts2_groupNUTS2 2
## 9  HQ_TTC TA Linear regression on log(HQ)              nuts2_groupNUTS2 3
## 10 HQ_TTC TA Linear regression on log(HQ)              nuts2_groupNUTS2 4
## 11 HQ_TTC TA Linear regression on log(HQ)              nuts2_groupNUTS2 5
## 12 HQ_TTC TA Linear regression on log(HQ)              nuts2_groupNUTS2 6
```

```
## 13 HQ_TTC TA Linear regression on log(HQ)             nuts2_groupNUTS2 7
## 14 HQ_TTC TA Linear regression on log(HQ)             occ_groupocc_s = 1
## 15 HQ_TTC TA Linear regression on log(HQ)         bmi_groupNormal weight
## 16 HQ_TTC TA Linear regression on log(HQ)            bmi_groupOverweight
## 17 HQ_TTC TA Linear regression on log(HQ)               bmi_groupObesity
## 18    HQ DON Linear regression on log(HQ)                    (Intercept)
## 19    HQ DON Linear regression on log(HQ)                       ageyears
## 20    HQ DON Linear regression on log(HQ)                  sex_groupMale
## 21    HQ DON Linear regression on log(HQ)              isced_groupMedium
## 22    HQ DON Linear regression on log(HQ)                isced_groupHigh
## 23    HQ DON Linear regression on log(HQ)       degurba_groupRural areas
## 24    HQ DON Linear regression on log(HQ) degurba_groupTowns and suburbs
## 25    HQ DON Linear regression on log(HQ)             nuts2_groupNUTS2 2
## 26    HQ DON Linear regression on log(HQ)             nuts2_groupNUTS2 3
## 27    HQ DON Linear regression on log(HQ)             nuts2_groupNUTS2 4
## 28    HQ DON Linear regression on log(HQ)             nuts2_groupNUTS2 5
## 29    HQ DON Linear regression on log(HQ)             nuts2_groupNUTS2 6
## 30    HQ DON Linear regression on log(HQ)             nuts2_groupNUTS2 7
## 31    HQ DON Linear regression on log(HQ)             occ_groupocc_s = 1
## 32    HQ DON Linear regression on log(HQ)         bmi_groupNormal weight
## 33    HQ DON Linear regression on log(HQ)            bmi_groupOverweight
## 34    HQ DON Linear regression on log(HQ)               bmi_groupObesity
##        estimate exp_estimate percent_change     p_value signif_label
## 1  -0.124482562    0.8829537    -11.7046342 0.897658087
## 2  -0.019769190    0.9804249     -1.9575061 0.365665863
## 3  -0.284321744    0.7525245    -24.7475514 0.066431841
## 4  -0.401957052    0.6690095    -33.0990522 0.039439715            *
## 5  -0.554509744    0.5743538    -42.5646228 0.004864649           **
## 6  -0.094285958    0.9100225     -8.9977503 0.654080032
## 7  -0.124264377    0.8831463    -11.6853674 0.551133134
## 8   0.119301253    1.1267093     12.6709291 0.679252387
## 9   0.317840881    1.3741576     37.4157590 0.215275504
## 10 -0.083900213    0.9195230     -8.0476992 0.793802820
## 11  0.419059034    1.5205301     52.0530115 0.116411963
## 12  0.218562686    1.2442870     24.4287014 0.362193485
## 13  0.014498724    1.0146043      1.4604340 0.957556030
## 14 -0.295562456    0.7441129    -25.5887061 0.280501450
## 15 -0.112167755    0.8938943    -10.6105711 0.835535690
## 16 -0.337882628    0.7132790    -28.6720998 0.537560916
## 17 -0.310964641    0.7327398    -26.7260215 0.578993898
## 18 -1.230392336    0.2921779    -70.7822077 0.360163591
## 19 -0.020199499    0.9800031     -1.9996856 0.505365604
## 20 -0.332174571    0.7173621    -28.2637919 0.122056435
## 21  0.198008988    1.2189733     21.8973349 0.463275883
## 22 -0.351921812    0.7033351    -29.6664888 0.195272171
## 23 -0.165844456    0.8471780    -15.2822003 0.570233667
## 24 -0.234581620    0.7909017    -20.9098322 0.417754110
## 25  0.603482685    1.8284757     82.8475729 0.132670107
## 26 -0.470544047    0.6246623    -37.5337670 0.186417749
## 27  0.001781242    1.0017828      0.1782829 0.996810390
## 28  0.109859620    1.1161214     11.6121378 0.766336026
## 29  0.633986020    1.8851097     88.5109708 0.057580633
## 30  0.382794657    1.4663769     46.6376889 0.311922421
## 31 -0.417234439    0.6588664    -34.1133562 0.272473359
```

```
## 32 -0.208920751    0.8114595    -18.8540459 0.780598307
## 33 -0.374465717    0.6876566    -31.2343416 0.622542421
## 34 -0.416745856    0.6591884    -34.0811573 0.592189787
```

```
##      HQ_type                          model                          term
## 1  HQ_TTC TA Logistic regression on HQ > 1                   (Intercept)
## 2  HQ_TTC TA Logistic regression on HQ > 1                      ageyears
## 3  HQ_TTC TA Logistic regression on HQ > 1                 sex_groupMale
## 4  HQ_TTC TA Logistic regression on HQ > 1             isced_groupMedium
## 5  HQ_TTC TA Logistic regression on HQ > 1               isced_groupHigh
## 6  HQ_TTC TA Logistic regression on HQ > 1       degurba_groupRural areas
## 7  HQ_TTC TA Logistic regression on HQ > 1 degurba_groupTowns and suburbs
## 8  HQ_TTC TA Logistic regression on HQ > 1             nuts2_groupNUTS2 2
## 9  HQ_TTC TA Logistic regression on HQ > 1             nuts2_groupNUTS2 3
## 10 HQ_TTC TA Logistic regression on HQ > 1             nuts2_groupNUTS2 4
## 11 HQ_TTC TA Logistic regression on HQ > 1             nuts2_groupNUTS2 5
## 12 HQ_TTC TA Logistic regression on HQ > 1             nuts2_groupNUTS2 6
## 13 HQ_TTC TA Logistic regression on HQ > 1             nuts2_groupNUTS2 7
## 14 HQ_TTC TA Logistic regression on HQ > 1            occ_groupocc_s = 1
## 15 HQ_TTC TA Logistic regression on HQ > 1         bmi_groupNormal weight
## 16 HQ_TTC TA Logistic regression on HQ > 1            bmi_groupOverweight
## 17 HQ_TTC TA Logistic regression on HQ > 1               bmi_groupObesity
## 18    HQ DON Logistic regression on HQ > 1                   (Intercept)
## 19    HQ DON Logistic regression on HQ > 1                      ageyears
## 20    HQ DON Logistic regression on HQ > 1                 sex_groupMale
## 21    HQ DON Logistic regression on HQ > 1             isced_groupMedium
## 22    HQ DON Logistic regression on HQ > 1               isced_groupHigh
## 23    HQ DON Logistic regression on HQ > 1       degurba_groupRural areas
## 24    HQ DON Logistic regression on HQ > 1 degurba_groupTowns and suburbs
## 25    HQ DON Logistic regression on HQ > 1             nuts2_groupNUTS2 2
## 26    HQ DON Logistic regression on HQ > 1             nuts2_groupNUTS2 3
## 27    HQ DON Logistic regression on HQ > 1             nuts2_groupNUTS2 4
## 28    HQ DON Logistic regression on HQ > 1             nuts2_groupNUTS2 5
## 29    HQ DON Logistic regression on HQ > 1             nuts2_groupNUTS2 6
## 30    HQ DON Logistic regression on HQ > 1             nuts2_groupNUTS2 7
## 31    HQ DON Logistic regression on HQ > 1            occ_groupocc_s = 1
## 32    HQ DON Logistic regression on HQ > 1         bmi_groupNormal weight
## 33    HQ DON Logistic regression on HQ > 1            bmi_groupOverweight
## 34    HQ DON Logistic regression on HQ > 1               bmi_groupObesity
##        estimate percent_change   p_value signif_label
## 1  3.789773e-02  -9.621023e+01 0.3476177
## 2  1.084460e+00   8.446028e+00 0.3629010
## 3  9.809644e-01  -1.903557e+00 0.9737014
## 4  7.396400e-01  -2.603600e+01 0.6246815
## 5  2.009810e-01  -7.990190e+01 0.0468867            *
## 6  1.930416e+00   9.304157e+01 0.3991470
## 7  8.012254e-01  -1.987746e+01 0.7928689
## 8  4.777611e-08  -1.000000e+02 0.9879308
## 9  1.147011e-01  -8.852989e+01 0.0677434
## 10 4.179812e-01  -5.820188e+01 0.5009193
## 11 5.557311e-01  -4.442689e+01 0.4961434
## 12 5.620481e-01  -4.379519e+01 0.4437306
## 13 3.762759e-01  -6.237241e+01 0.2940214
## 14 2.026316e+00   1.026316e+02 0.5964726
```

```
## 15 2.164483e-01  -7.835517e+01 0.3132565
## 16 1.245057e-01  -8.754943e+01 0.1803758
## 17 1.955499e-01  -8.044501e+01 0.2977796
## 18 2.352818e-21  -1.000000e+02 0.9996829
## 19 3.069377e-01  -6.930623e+01 0.2165487
## 20 7.500145e+06   7.500144e+08 0.9991353
## 21 1.234127e-15  -1.000000e+02 0.9989888
## 22 7.288478e+05   7.288468e+07 0.9994721
## 23 7.007151e-21  -1.000000e+02 0.9976865
## 24 3.571726e-28  -1.000000e+02 0.9978483
## 25 9.342668e+01   9.242668e+03 0.9999275
## 26 4.883619e-01  -5.116381e+01 0.9999872
## 27 3.378536e-01  -6.621464e+01 0.9999854
## 28 3.384152e+30   3.384152e+32 0.9985342
## 29 2.610447e+21   2.610447e+23 0.9989048
## 30 1.060871e+00   6.087071e+00 0.9999990
## 31 4.691417e-04  -9.995309e+01 0.2676937
## 32 1.316891e+16   1.316891e+18 0.9997430
## 33 7.150482e+06   7.150481e+08 0.9998904
## 34 8.204193e-05  -9.999180e+01 0.9999363
```

Save outputs

This chunk creates an `outputs` folder and saves all summary tables and figures generated in the report in one place.